Fake News Detector — Project Report

Oct 21, 2026 · @Vishal Senapati

Problem Statement

Who has the problem? Everyday social media users, students, journalists, and institutions who consume news online — especially on platforms like WhatsApp, Facebook, Twitter/X, and Instagram, where content spreads fast without verification.

What exactly is happening? False or misleading news, fabricated headlines, and out-of-context content are shared widely and quickly, often mixed in with legitimate news, making it hard for an average reader to tell them apart.

Why is it a problem? People don't have the time, tools, or media literacy to fact-check every piece of content they see. Fake news is often designed to look credible (fake logos, doctored screenshots, sensational headlines), so it spreads faster than the truth — false news has been shown to spread significantly faster than true news on social media.

What is the impact?

- Social: communal tension, panic (e.g. fake health advisories), mob violence triggered by rumors
- Political: election misinformation, manipulated public opinion
- Personal: financial scams, wrong health decisions, reputational damage
- Institutional: loss of trust in genuine media and journalism

How is it handled currently?

- Manual fact-checking organizations (e.g. Alt News, BOOM, PolitiFact)
- Platform-level flagging (community notes, warning labels)
- Government advisories and takedown requests
- User reporting mechanisms on social platforms

Why is the current approach insufficient?

- Manual fact-checking is slow — by the time something is verified, it has already spread to millions
- Fact-checkers can't keep up with the volume of content generated every second
- Platform flagging is inconsistent and reactive, not preventive

- Most solutions target English/major languages, leaving regional language content largely unchecked
- Users still have to rely on someone else's judgment; there's no accessible tool for instant, on-the-spot verification

The gap: an automated, fast, accessible first-line filter that gives users an instant credibility signal — not replacing fact-checkers, but reducing the volume of unchecked content and buying time before things go viral.

Objective

To design and build an AI-powered platform that gives users an instant, reliable credibility signal for news content — reducing the spread of misinformation before it goes viral, while keeping every verification result tamper-proof.

Specific objectives:

1. Classify news text as credible or fake using ML/DL models (TF-IDF baseline → LSTM → transformer-based)
2. Score source credibility using domain reputation and publishing history
3. Detect manipulated images via reverse-image and metadata checks
4. Record every verification result immutably on a blockchain audit log
5. Support regional languages (e.g. Hindi) alongside English
6. Deliver results through a simple web app / browser extension

Methodology

The baseline follows a standard text-classification pipeline:

1. Data cleaning: lowercase, remove punctuation and stopwords, drop nulls/duplicates
2. Feature extraction: TF-IDF vectorization of article title + body text
3. Model training: Logistic Regression and Naive Bayes trained as baseline classifiers, compared on the same 80/20 train/test split
4. Evaluation: accuracy, precision, recall, F1-score, and confusion matrix
5. Deployment: a Streamlit app takes raw text input, runs it through the trained vectorizer + model, and returns a Real/Fake prediction with a confidence score

Future phases (see Next Steps) add deep learning models (LSTM, transformer-based), source credibility scoring, and a blockchain audit log on top of this baseline.

Dataset

To be added — see dataset selection notes: Kaggle Fake News Dataset (Phase 1 baseline), ISOT Fake News Dataset (Phase 2 scale-up), LIAR Dataset (optional stretch), FakeNewsNet (Phase 3, source credibility).

Phase 1 training used the Kaggle Fake News Dataset to build and validate the baseline model. *[Add: number of rows used, class balance, any cleaning steps specific to this dataset.]* Metrics and the confusion matrix for this run are captured in the Results section below.

Results

Baseline model performance on the held-out test set:

- Model used: Logistic Regression (best performer)
- Accuracy: 98.82%
- Precision: 0.9882
- Recall: 0.9882
- F1-score: 0.9882
- Confusion matrix: *[insert screenshot/image here]*

Logistic Regression outperformed Naive Bayes across every metric (98.82% vs. 92.8% accuracy, precision, recall and F1-score), so it was selected as the baseline model going forward. Naive Bayes is still a solid, fast baseline and is noted here for comparison.

Challenges Faced

- Environment setup (Python not on PATH, pip/streamlit not recognized) took longer than planned and ate into Day 1 time
- *[Add any data cleaning issues faced, e.g. class imbalance, missing/duplicate rows]*
- *[Add any model training issues, e.g. overfitting, slow training]*
- *[Add any issues integrating the trained model into the Streamlit app]*

Next Steps

- Phase 2: upgrade from TF-IDF + Logistic Regression to LSTM, then fine-tune a transformer model (BERT/DistilBERT) for higher accuracy

- Phase 3: add source credibility scoring, image manipulation detection, and regional language (Hindi) support
- Phase 4: build the blockchain-based immutable audit log for every verification result
- Phase 5: polish the web app/browser extension, add a dashboard, complete testing and documentation